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ELECTRICAL POWER GENERATION

Introduction.

The grid comprises of power stations throughout the country. These stations feed current into a network of conductors at a very high voltage. Overhead conductors, supported by pylons (steel towers), are used to transmit power over long distances so as to reach consumers.

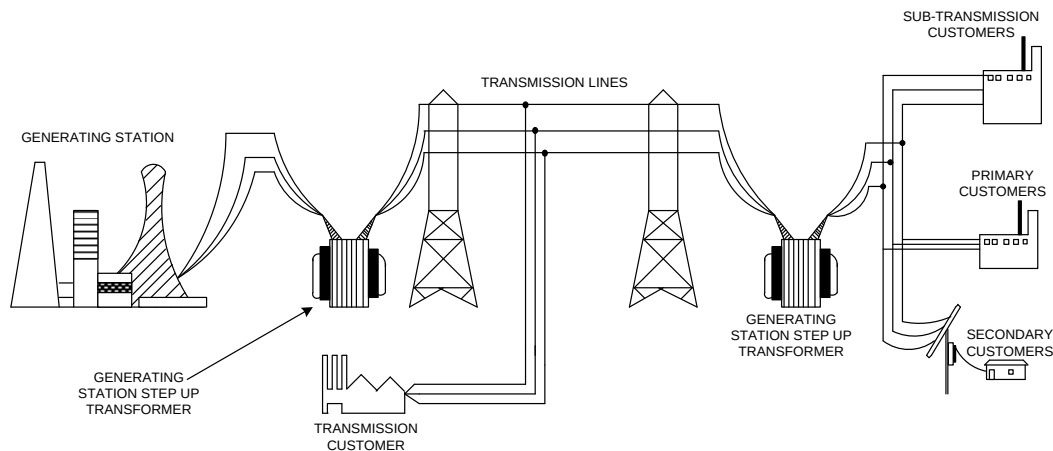


Fig.8.01 Generation, Transmission and distribution

AC Power Supply system

The network between Generating Station and consumers of electric power can be divided into two.

- i. *Transmission System*
- ii. *Distribution System.*

Transmission can further be divided into *Primary transmission and secondary transmission*. Similarly distribution can also be divided into *primary distribution and secondary distribution*.

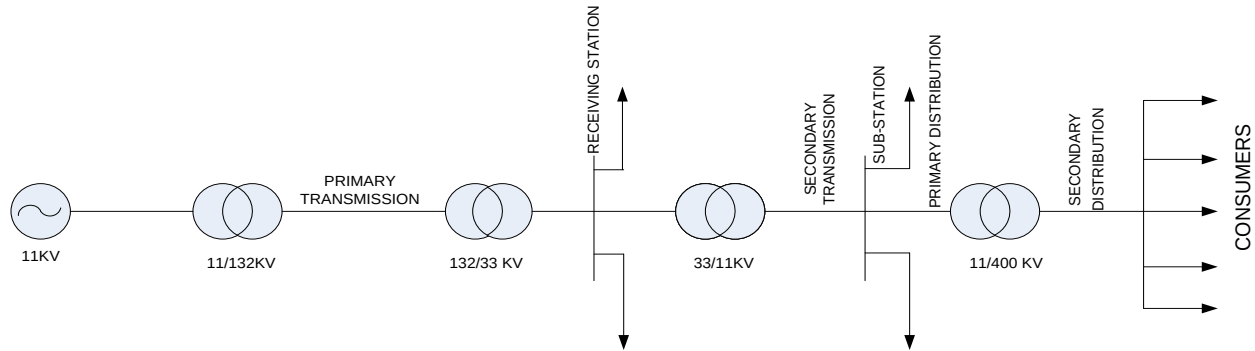


Fig 8.02 Single Line diagram of AC power System

1. **Generating Station**
2. **Primary transmission**
3. **Secondary transmission**
4. **Primary Distribution**
5. **Secondary Distribution**

Generating Station:

The place where electric power is generated by three phase generators. The generated voltage is 11kV. The voltage is then stepped up to 132kV by use of three phase transformers for transmission.

Primary Transmission:

The stepped up voltage at 132KV is now transmitted to load centers by three phase 3-wire overhead system.

Secondary transmission:

Primary transmission terminates at a receiving station where voltage is stepped down to 33KV by use of 3-phase step-down transformers. From the station electric power is transmitted at 33KV by use of 3-phase step-down transformers. This is for areas far away from generating stations.

Primary Distribution:

Secondary transmission terminates at a sub-station where voltage is stepped down from 33KV to 11KV by three phase transformers. The 11KV line runs around load centers by three-phase 3-wire forming primary distribution. Heavy consumers with demands of more than 50KW are supplied power at 11KV for further handling with their own sub-stations.

Secondary Distribution:

The electric power from primary distribution (11kV) is delivered to distribution sub-stations, these sub-station are located near the consumers areas where the level of voltage is reduced by step-down transformers to 400V three phase 4-wire for secondary distribution. So there is 400 Volts (Three Phase Supply System) between any two phases and 230 Volts (Single Phase Supply) between a neutral and phase (live) wires. Residential load may be connected between any one phase and neutral wires, while three phase load may be connected directly to the three phase lines.

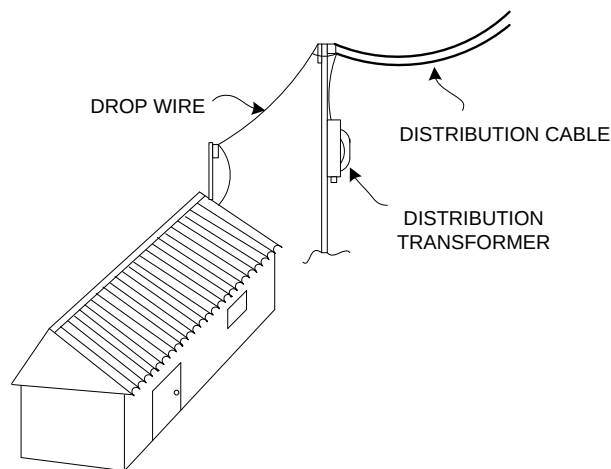


Fig 8.03 supply to consumer

Elements of Distribution System

Secondary distribution may be divided into three parts

1. **Feeders**
2. **Distributors**
3. **Service Lines or Service Mains**

Feeders:

Those Electric lines which connect Generating station (power station) or Sub Station to distributors are called feeders.

Distributors:

Those tapping which extracted for supply of electric power to the consumers or those lines, from where consumers get electric supply are called distributors. The difference between Feeder and Distributor is that Current in Feeder is same throughout while Voltage is same in each section of Distributor

Service Lines or Service Mains:

The cable connected between Distributors and Consumer load terminal called Service Line or Service Mains.

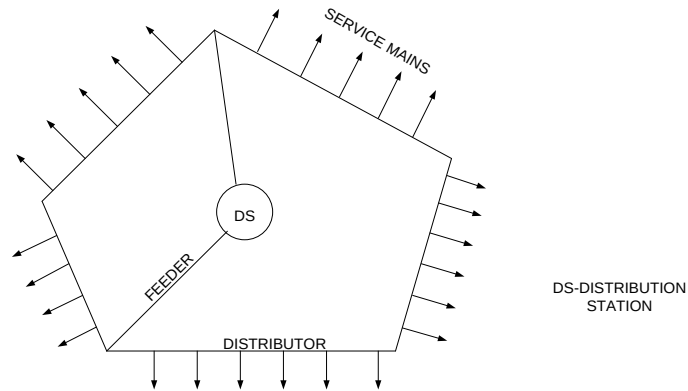


Fig 8.04 elements of low voltage distribution system

AC GENERATION

Generator Construction

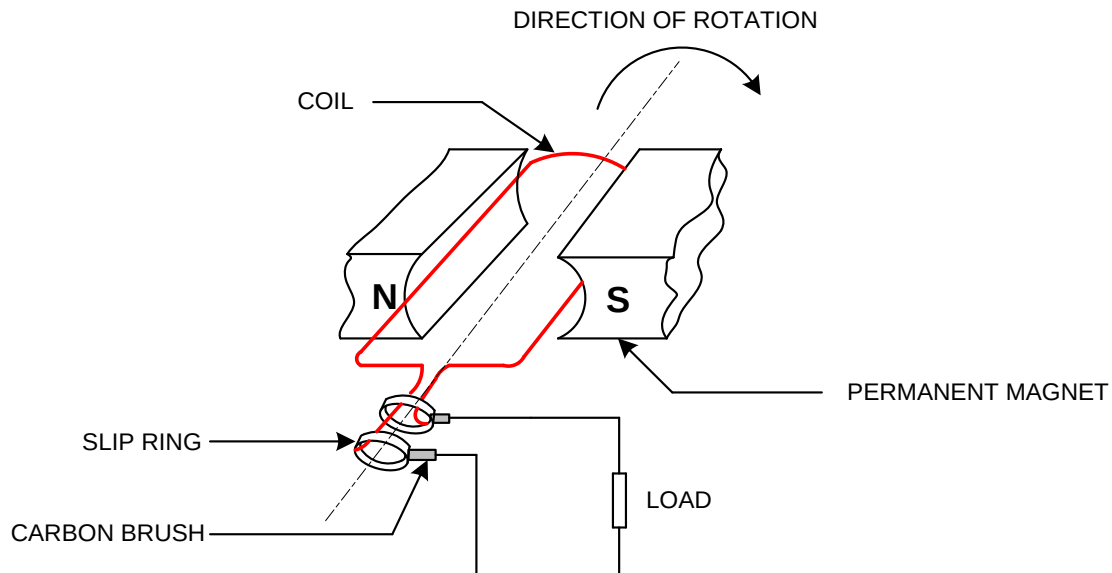


Fig 1.01 Simple AC generator

The generator consists of the following parts

- (a) Permanent magnet of two poles that produces stationary magnetic field.
- (b) A coil of conductor that is rotated in the magnetic field causing a voltage to be induced in it.
- (c) Two slip rings connected to the two ends of the coil from which the generated voltage can be tapped and supplied to the load.
- (d) Carbon brushes that are spring loaded attached to the slip rings that conduct current to the load.
- (e) Conductors connected to the carbon brushes and the external load. They provide a path for current flow between the coil and the load.

PRINCIPLE OF OPERATION OF AN A.C. GENERATOR

The basic ac generator shown in fig.1.01 is a pivoted coil of sides AB and CD which rotates between the two pole pieces of a magnet with a uniform speed in a uniform stationary magnetic field. As the coil rotates, it cuts the lines of flux, an emf. is induced in the conductor because of relative motion between the field and the conductor. The direction of induced emf. is given by Flemings Right Hand Rule as shown in fig 1.02

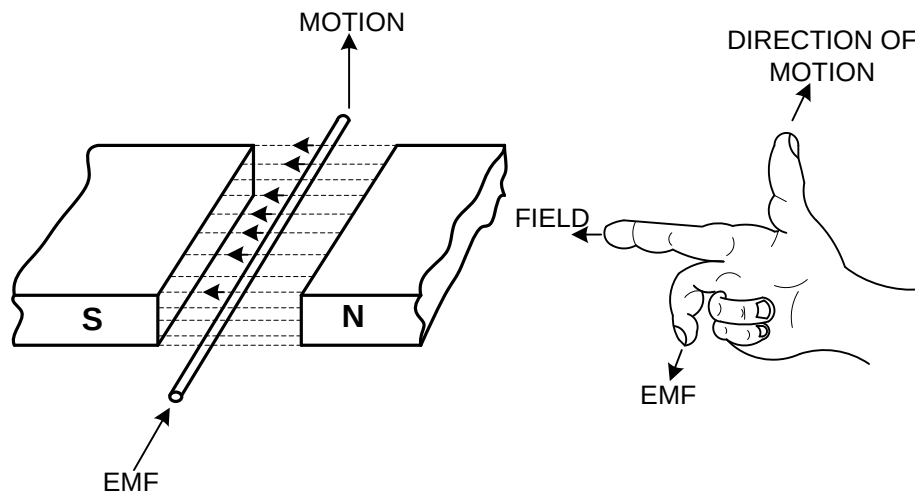


Figure 1.02 Flemings Right Hand Rule

In fig 1.03 the conductor AB which is rotating in an anticlockwise direction is at a position of cutting no flux lines, the induced emf. will be zero.

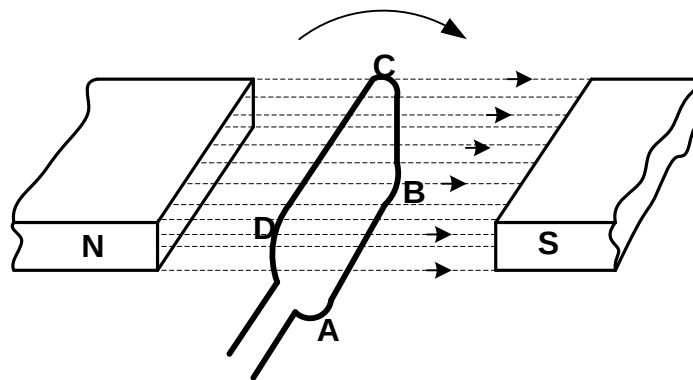


Figure 1.03 Conductor at a position cutting no magnetic lines of force

When the conductor reaches the position shown in fig.1.03 maximum e.m.f. will be induced in conductor.

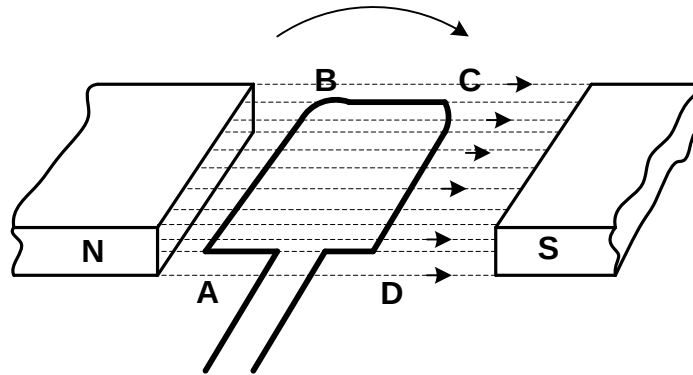


Fig 1.04 Conductor at a position cutting maximum magnetic lines of force

As the conductor moves to position shown in figure 1.04 the emf. induced in the coil is again zero since the conductor is parallel to the magnetic lines of flux.

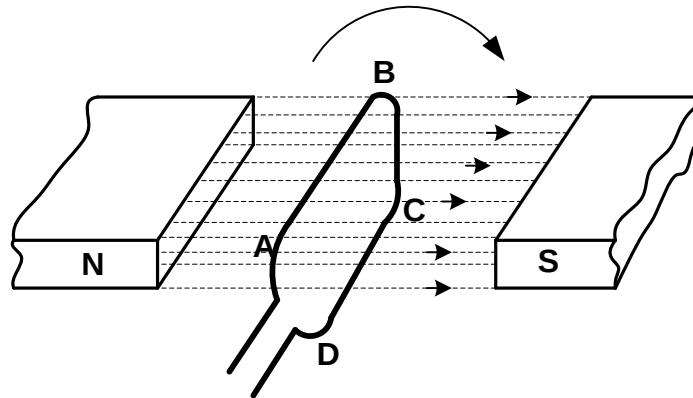


Fig 1.05 Conductor at a position cutting no magnetic lines

When the conductor AB moves to the position shown in figure 1.05, maximum emf is induced in it. The direction of the induced emf. will be found to be reversed applying Flemings Right Hand Rule.

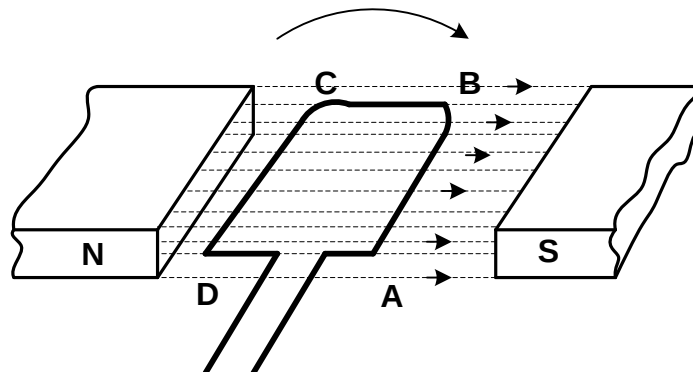


Fig 1.06. Conductor at a position cutting maximum magnetic lines of force

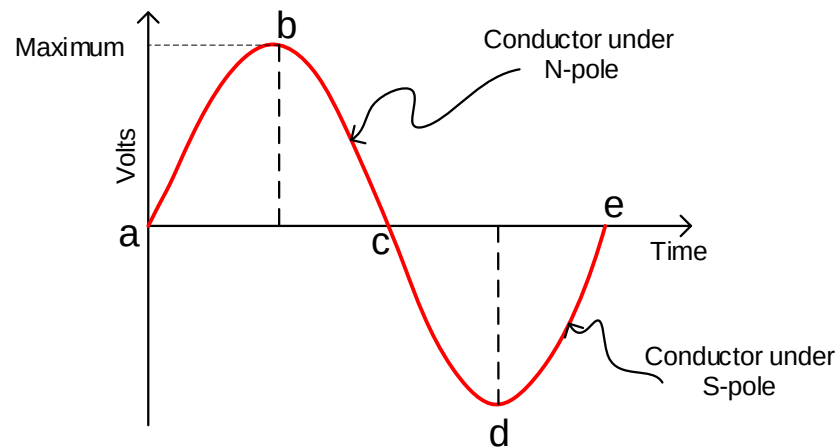


Fig 1.07. Generated emf.

The generated e.m.f. is an alternating voltage that varies in *magnitude* and *direction*.

Figure 1.06 above shows a waveform representing one complete set of changes in an alternating voltage. The figure shows the magnitude and direction of voltage in a single conductor rotating through 360° (one complete revolution) between the poles of the magnet.

Voltage variations in one revolution:

Position fig. 1.03: The conductor is not cutting any lines of flux and therefore the voltage is zero.

Position fig. 1.04: The conductor is cutting lines of flux at right angles and therefore producing maximum voltage.

After *position fig. 1.05*, the conductor is cutting lines of flux in the opposite direction and therefore the voltage is in the opposite direction.

The generated voltage is a varying quantity that generates a sign wave as shown below.

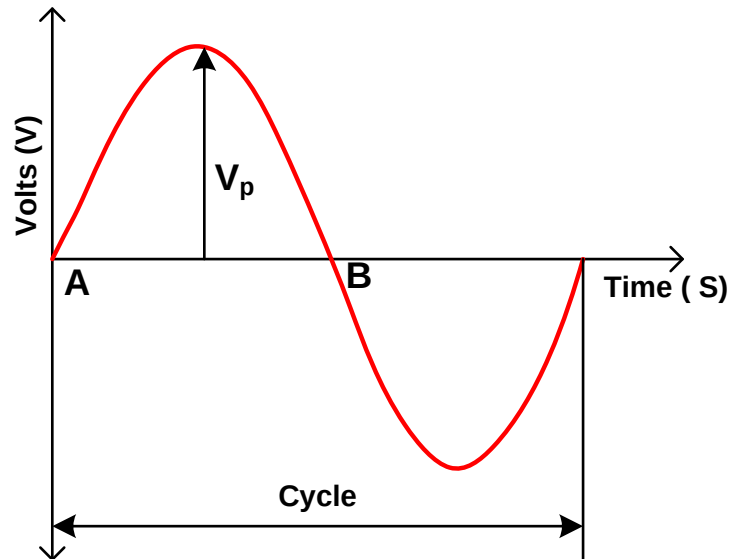


Figure 1.08 sine wave

Terms associated with the sine wave

Cycle

A cycle is one complete set of changes that are repetitive. In figure 1.07 the cycle starts from A to C.

Instantaneous value

This is the value of the alternating voltage at any given time.

Peak value

This is the maximum value of the alternating voltage

Average value

This is the arithmetical average of an alternating quantity over one cycle.

Root Mean Square (R.M.S.) value

This is the value of alternating voltage or current that will give the same heating effect as a steady voltage or current (D.C.) when passing through the same resistance.

ELECTRICITY GENERATING STATIONS.

A generating station comprises a prime mover which drives an alternator for the production of electric power. The prime mover (turbine), converts energy from some other form into mechanical energy. The alternator converts mechanical energy of the prime mover into electrical energy. The electrical energy produced by the generating station is transmitted and distributed with the help of conductor network to various consumers. Depending upon the form of energy converted into electrical energy, the generating stations are classified as follows:

- i. Hydroelectric power stations
- ii. Steam power stations
- iii. Diesel power stations
- iv. Nuclear power stations.
- v. Gas turbine power plant.

Major generating stations in Kenya

Hydro-Electric power stations.

These include; The Seven Forks Scheme which generates almost all of Kenya's Hydro-electric power. It generates approximately 530 MW of power. The scheme comprises the following stations;

- (i) Masinga Power Station
- (ii) Gitaru Power Station
- (iii) Kamburu Power Station
- (iv) Kindaruma Power Station
- (v) Kiambere Power Station

Gitaru is the biggest power station in Kenya in terms of installed capacity.

Geothermal Stations

Kenya is determined to make Geothermal the main power source in the country, in the Great Rift Valley, Kenya has built four plants to exploit the geothermal resource;

- (i) Olkaria I (185 MW),
- (ii) Olkaria II (150 MW)
- (iii) Olkaria III which is privately owned plant (110 MW).
- (iv) Olkaria IV (140 MW)

Kenya is still continuing to explore and exploit this resource.

Wind power

- (i) *Ngong Hills Wind Farm* is currently the only major wind farm; It is located in Ngong, Kajiado County and produces about 5.1 MW of electricity. It is owned by Kenya Electricity Generating Company (KenGen).
- (ii) *Lake Turkana Wind Power (LTWP)* Kenya plans to build the largest wind farm in Africa, which aims to provide 300MW.

Independent Power Producers (IPPs)

These contributes to about 10% of the total power

- Westmont
- Iberafrica
- OrPower4 (Kenya) subsidiary of Ormat Technologies
- Tsavo Power Company (TPC)
- Aggreko
- Africa Geothermal International (AGIL)

HYDRO-ELECTRIC POWER STATION.

Hydro-electric power stations are located in hilly areas where dams can be built and large water reservoirs can be obtained. In a hydroelectric power station, water head is created by constructing a dam across a river or a lake. From the dam, water is fed to a water turbine. The water turbine converts the energy in the falling water into mechanical energy at the turbine shaft. The turbine drives the alternator which in turn converts mechanical energy into electrical energy.

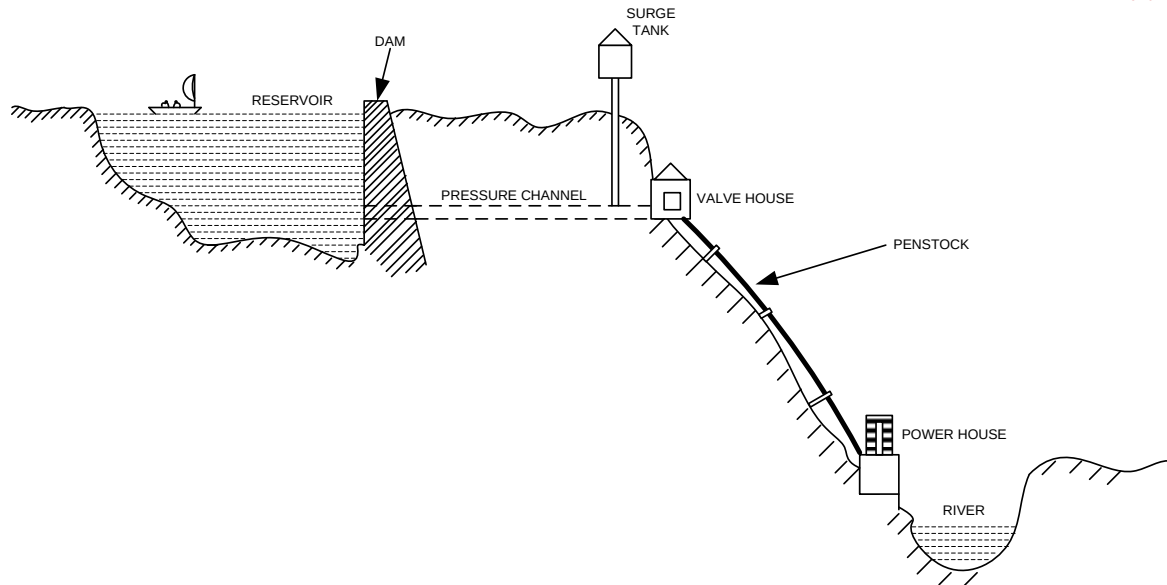


Fig 8.02 Schematic arrangement of a Hydro- electric plant

The constituents of a hydro-electric plant are (1) hydraulic structures (2) water turbines and (3) electrical equipment.

Hydraulic structures. Hydraulic structures in a hydro-electric power station include dam, spillways, head works, surge tank, penstock and accessory works.

- (i) *Dam.* A dam stores water and creates water head. The type and arrangement depends upon the topography of the site.
- (ii) *Spillways.* Spillways are used to discharge surplus water from the storage reservoir into the river on the down-stream.
- (iii) *Head works.* These consist of the diversion structures at the head of an intake. They generally include booms and racks for diverting floating debris.
- (iv) *Surge tank.* When closed conduits are used always with a surge tank provided. A surge tank which is open at the top reduces the pressure swings in the conduit when water level rises or falls to.
- (v) *Penstock .*penstock is open or closed conduits which carry water to the turbines.

Electrical equipment. The electric equipment of a hydroelectric power station includes alternators, transformers, circuit breakers and other switching and protective devices.

Turbine. The water turbine captures the energy in the falling water and changes the hydraulic energy into mechanical energy at the turbine shaft.

Factors to consider when choosing the site for Hydro-electric plant

- (i) ***Availability of water***
- (ii) ***Storage of water***; that site selected for a hydro-electric plant should provide adequate facilities for erecting a dam and storage of water.
- (iii) ***Cost and type of land***. The land for the construction of the plant should be available at a reasonable price, further the bearing capacity of the ground should be adequate to withstand the weight of heavy equipment to be installed.
- (iv) ***Transportation facilities***. The site selected for a hydro-electric plant should be accessible by rail and road so that necessary equipment and machinery could be easily transported.

Advantages

- (i) It requires no fuel as water is used for the generation of electrical energy
- (ii) It is quite neat and clean as no smoke or ash is produced
- (iii) Since water is the source of energy, It requires very small running costs.
- (iv) It is simpler in construction and requires less maintenance.
- (v) It does not require a long starting time.
- (vi) It is robust and has a longer life.
- (vii) It improves the area in agricultural activities through irrigation and controlling floods.
- (viii) Requires few experienced personnel when running.

Disadvantages

- (i) Highly skilled personnel required to build the plant.
- (ii) High capital cost due to construction of dam.
- (iii) There is uncertainty about the availability of huge amount of water due to dependence on weather conditions.
- (iv) Cost of transmission is high since the plant is located in hilly areas far away from the consumer.

STEAM POWER STATION (THERMAL STATION).

This generating station converts heat energy from coal combustion into electrical energy. Steam is produced in a boiler by utilizing the heat of coal combustion. The steam is then expanded in the prime mover (steam turbine) and is condensed in a condenser and sent back to the boiler again.

The turbine drives the alternator which converts mechanical energy of the turbine into electrical energy. This type of power station is suitable where coal and water are abundantly available.

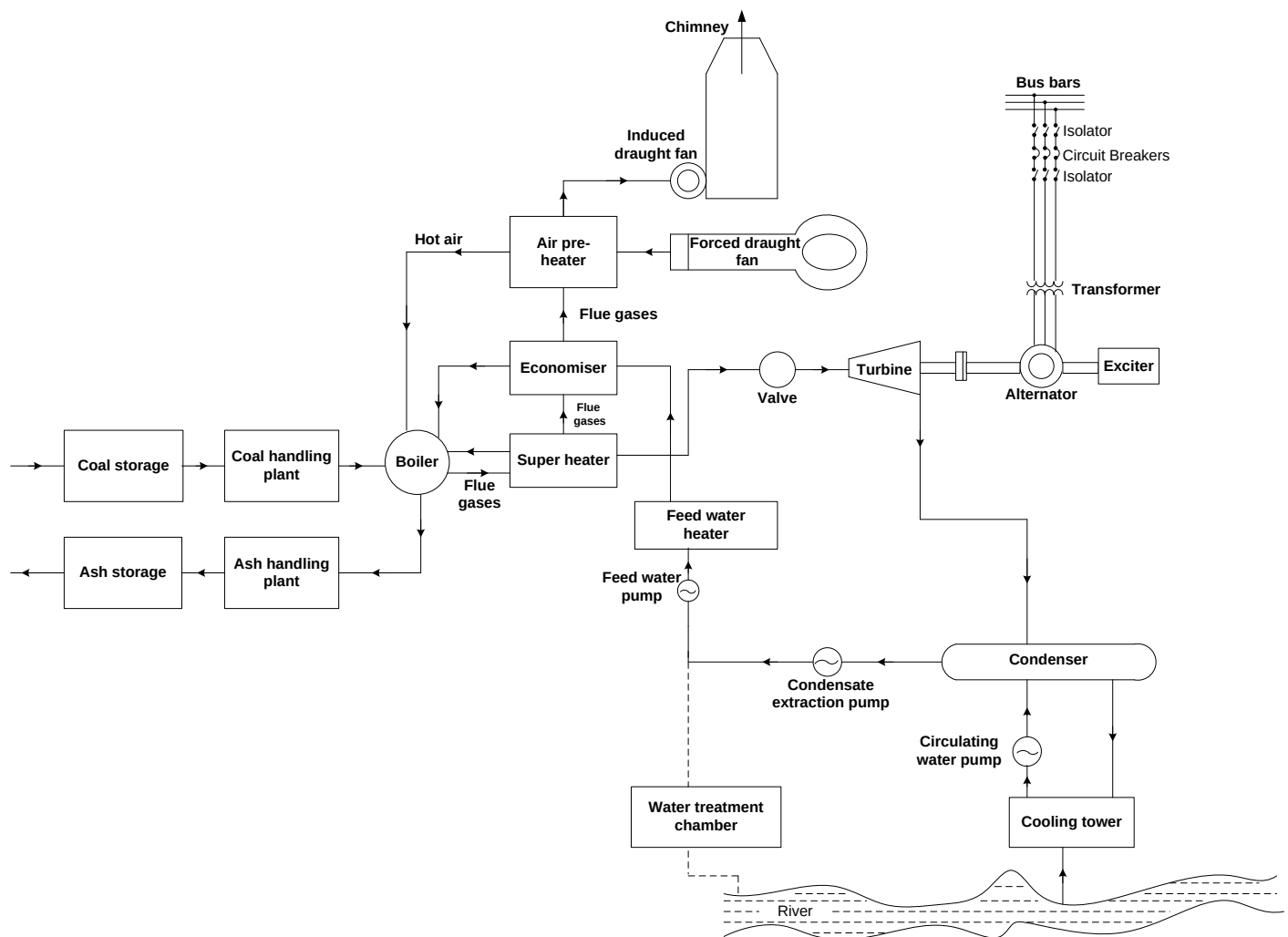


Fig 8.03 Schematic arrangement of Steam Power Station

The station consists of the following

1. Coal and ash handling arrangement
2. Steam generating plant
3. Steam turbine
4. Electrical equipment
5. Water treatment plant
6. Cooling arrangement

Coal and ash handling plant.

Coal is transported to the power station by road or by rail and stored in the coal storage plant. From the coal storage plant, coal is delivered to the handling plant where it is crushed into small pieces in order to increase its surface area, to improve. The crashed coal is fed to the boiler by conveyors. The coal is burnt in the boiler and the ash produced after complete combustion of coal is removed to the ash handling plant and then disposed off. The removal of the ash from the boiler facilitates proper burning of coal.

Steam generating plant.

The steam generating plant consists of a boiler for the production of steam and other auxiliaries for the utilization of flue gases, it consists of;

- (i) *Boiler*; this is closed vessel in which water is converted into steam through heating by coal combustion. The boilers are broadly classified into:
 - (a) Water tube boilers
 - (b) Fire tube boilers

In a *water tube boiler*, water flows through the tubes and the hot gases of combustion flow over these tubes

In a *fire tube boiler*, the hot products of combustion pass through the tubes surrounded by water. the heat of combustion of coal in the boiler is utilized to convert water into steam at high temperature and pressure. The flue gases from the boiler passes through; super heater, economizer, and air pre-heater and are finally exhausted to atmosphere through the chimney.

- (ii) *Super heater*; A super heater is a device which superheats the steam raising the temperature of steam above boiling point of water. This increases the overall efficiency of the plant. A super heater consists of a group of tubes made of special alloy steels. These tubes are heated by the heat of flue gases during their journey from the furnace to the chimney. The steam produced in the boiler is led through the super heater where it is superheated by the heat from the flue gases. Super heaters are mainly classified into two types according to the system of heat transfer from flue gases to steam ;
 - (a) Radiant super heater
 - (b) Convection super heater.

The radiant superheated is placed in the furnace between the water walls and receives heat from the burning fuel through radiation process.

The convection super heater is laced in the boiler tube bank and receives heat from flue gases entirely through the convection process. This type of super heater is commonly used these days.

(iii) *Economizer*; the economizer derives heat from the flue gases to heat feed of water. The feed water is fed to the economizer before heading to the boiler. The economizer extracts a part of heat of flue gases to increase the feed water temperature.

(iv) *Air-preheater*; An air preheater increases the temperature of the air supplied for coal burning by deriving heat from coal gases. Air is drawn from the atmosphere by forced draught fan and passed through air preheater before supplying to the boiler furnace. The air preheater extracts heat from flue gases and increases the temperature of air used for coal combustion.

Steam turbine.

The dry and superheated steam from the super heater is fed to the steam turbine through main valve. The heat energy of steam when passing over the blades of turbine is converted into mechanical energy.

Electrical equipment.

These include:

- i. *Alternator*. Each alternator is coupled to a steam and converts mechanical energy of the turbine into electrical energy.
- ii. *Transformer*. A generating station has different types of transformers.
 - (a) Main step-up transformer which step-up the generation voltage for transmission of power.
 - (b) Station transformer which are used for general services like lighting the power station.
 - (c) Auxiliary transformer which supply to individual unit-auxiliaries.
- iii. *Switchgear*. The equipment at switchgear locates the faulty part on the system and isolates it from the healthy section. The gear consists of circuit breakers, relays, switches and other control devices.

Water treatment plant.

Boilers require clean and soft water. The source of boiler feed water is generally a river or lake which may contain impurities .The water from the source of supply is stored in storage tanks

and there after purified and softened by chemical treatment and then delivered to the boiler. The pure and soft water available is fed to the boiler for steam generation.

Cooling arrangement.

To improve the efficiency of the plant, steam exhausted from the turbine is condensed by means of a condenser. Water is circulated through the condenser. The circulating water takes up the heat of the exhausted steam and the pump is heated. cooling towers are added to enhance cooling. The cold water from the cooling tower is reused in the condenser.

Factors to consider when selecting a site for steam power station:

- (i) *Availability of water.*
- (ii) *Supply of fuel.*
- (iii) *Transportation facilities.*
- (iv) *Cost and type of land.*
- (v) *Nearness to load centers.*
- (vi) *Distance from populated area.*

Advantages

- i. The fuel used is quite cheap.
- ii. Low initial cost.
- iii. It can be installed at anywhere. The coal can be transported to the site of the plant by rail or road.
- iv. Requires less space as compared to the hydroelectric power station.
- v. Less generation cost than diesel power station.

Disadvantages

- i. It pollutes the atmosphere due to the production of smoke and fumes.

DIESEL POWER STATION

This is a station in which diesel engine is used as the prime mover for the generation of electrical energy is known as diesel station.

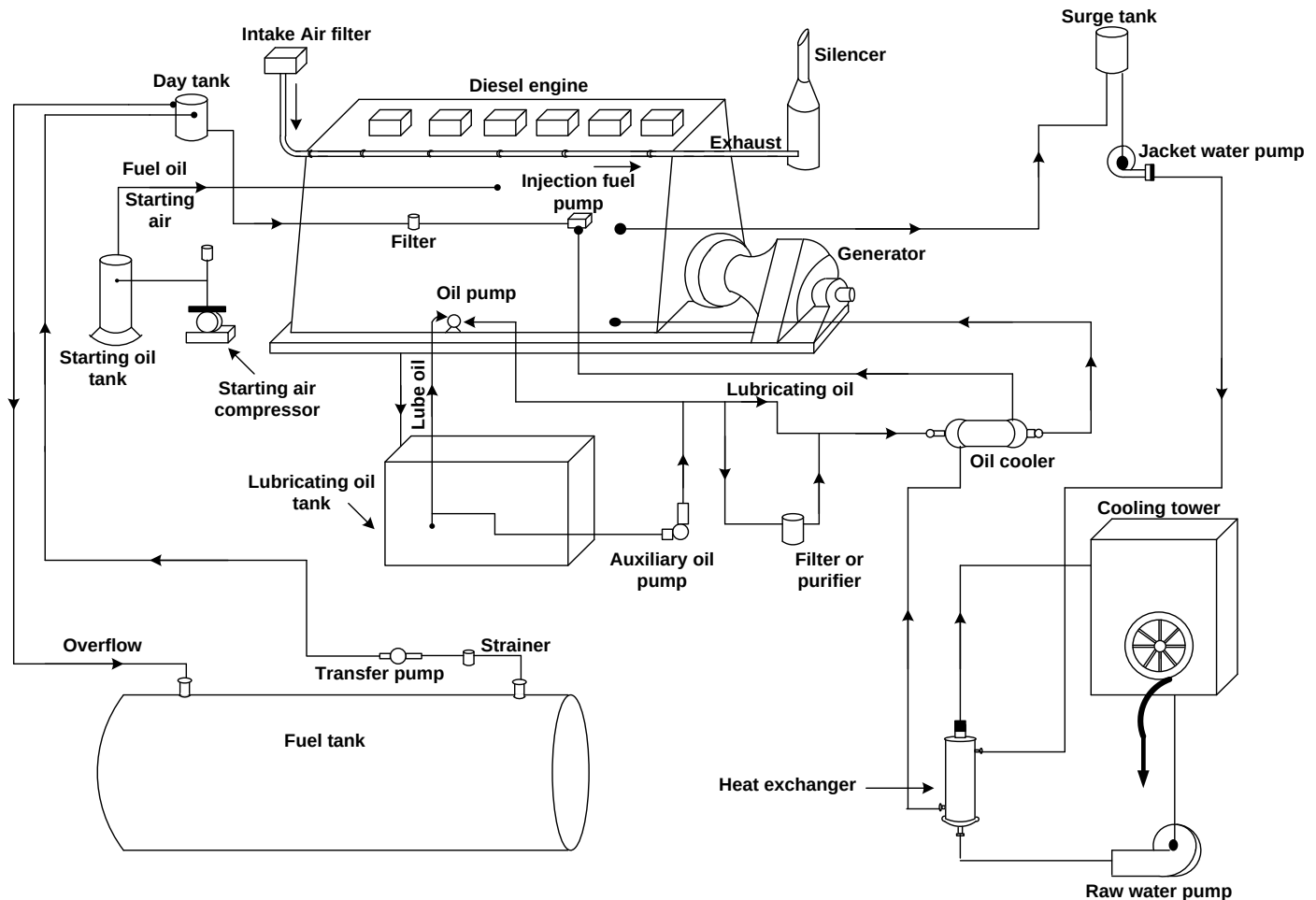


Fig 8.04 Schematic arrangement of Diesel Power plant

In addition to diesel generator set, the plant has the following auxiliaries:

- (i) Fuel supply system.
- (ii) Air intake system.
- (iii) Exhaust system.
- (iv) Cooling system.
- (v) Cylinder and head jacket.
- (vi) Lubricating system.
- (vii) Engine starting system.

Here diesel engine is used as the prime mover. The diesel burns inside the engine and the product of this combustion produces mechanical energy. The engine drives the alternator hence converting mechanical energy into electrical energy.

Advantages

- (i) There are no standby losses.
- (ii) The design and layout is simple
- (iii) It occupies less space
- (iv) It can be located at any place
- (v) It requires less quantity of water for cooling
- (vi) The overall cost is much less than that of steam power station.
- (vii) It requires less operating staff.

Disadvantages

- (i) The cost of lubrication is high.
- (ii) The maintenance charges are high.
- (iii) The plant has high running charges.
- (iv) The plant can only generate small power.

NUCLEAR POWER STATION.

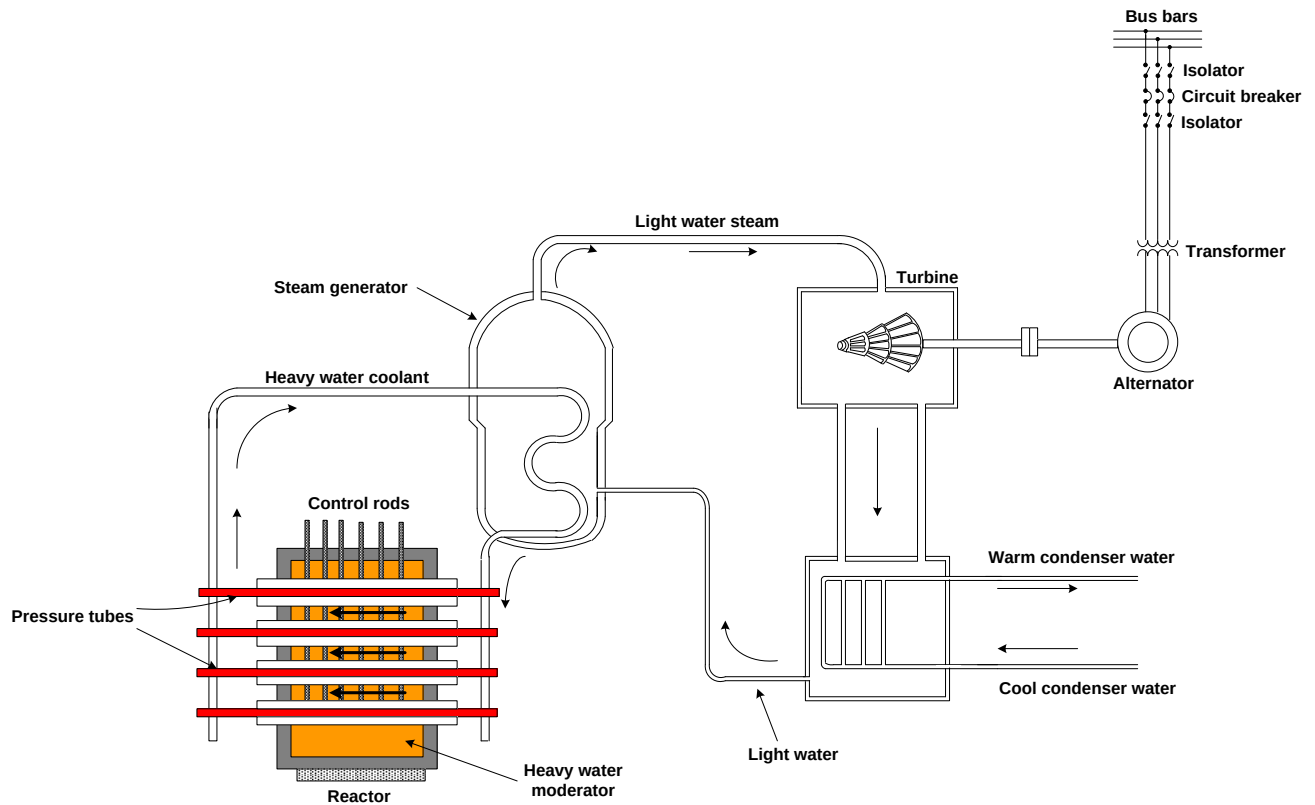


Fig 8.05 Schematic arrangement of nuclear power station

A generating station in which nuclear energy is converted into electrical energy is known as a nuclear power station.

The whole plant can be divided into the following main stages:

- (i) Nuclear reactor
- (ii) Heat exchanger
- (iii) Steam turbine
- (iv) Alternator

Nuclear reactor

This is a cylindrical pressure vessel which houses;

- a) Fuel rods (Uranium)
- b) Moderator and
- c) Control rods.

The *fuel rods* constitute the fission material and releases energy when bombarded with slow moving neutrons.

The *moderator* which consists of graphite rods slows down the neutrons before they bombard the fuel rods.

The *control rods* are of cadmium and are inserted into the reactor.

Cadmium is strong neutron absorber and regulates the supply of neutrons for fission. When the control rods are pushed in deep enough, they absorb most of fission neutrons and hence few are available for chain reaction, stopping the reaction. When the control rods are pulled out, more of the fission neutrons cause fission and enhance the chain reaction increasing heat production. Therefore, by pulling out the control rods, power of the nuclear reactor is increased, whereas by pushing them in, it is reduced. The pushing in and pulling out of the control rods is automatic and follows the load demand. The heat produced in the reactor is removed by coolant. The coolant carries the heat to the heat exchanger where the heat is dissipated.

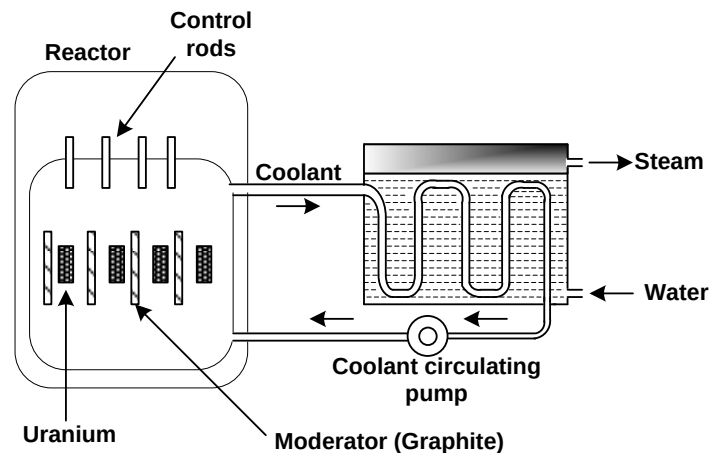


Fig 8.06 Nuclear Reactor

The Heat exchanger, Steam turbine and Electrical equipment are the same as in thermal station above.

Factors to consider when selecting the site for a nuclear power station.

- (i) *Availability of water.*
- (ii) *Disposal of waste.*
- (iii) *Distance from populated areas.*
- (iv) *Transportation facilities.*

Advantages

- (i) The amount of fuel required is quite small and therefore very economical.
- (ii) The plant requires less space.
- (iii) Nuclear deposits are readily available all over the world.
- (iv) It can be located near the load centers hence reducing the cost of primary distribution.
- (v) It quite reliable in operation.

Disadvantages

- (i) The fission by-products are generally radioactive and may cause a dangerous amount of radioactive pollution.
- (ii) The erection and commissioning of the plant requires highly skilled manpower.
- (iii) High Capital cost.

GAS TURBINE POWER PLANT.

The plant employs gas turbine as the prime mover. In this power plant, air is used as the working fluid. The air is compressed by a compressor and led to the combustion chamber where heat is added to air raising its temperature. Heat is added to the compressed air by use of air heaters. The hot and high pressure air from the combustion chamber is then passed to the gas turbine where it expands and turns the gas turbine. The gas turbine then drives the alternator which converts mechanical energy into electrical energy.

ARRANGEMENT OF GAS TURBINE POWER PLANT.

The components of this plant include:

- (i) Compressor
- (ii) Regenerator
- (iii) Combustion chamber
- (iv) Gas turbine
- (v) Electrical equipment
- (vi) Starting motor.

Compressor. The air from atmosphere pressure is sucked the compressor. The air is filtered to remove dust. The compressor pushes the air between stationary blades to raise its pressure. Hence air is available at high pressure at the output of the compressor.

Regenerator. A regenerator is a device which recovers heat from the exhaust gases of the turbine. The compressed air from the compressor passes through tubes to the combustion chamber. The compressed air is therefore heated by the hot exhaust gases.

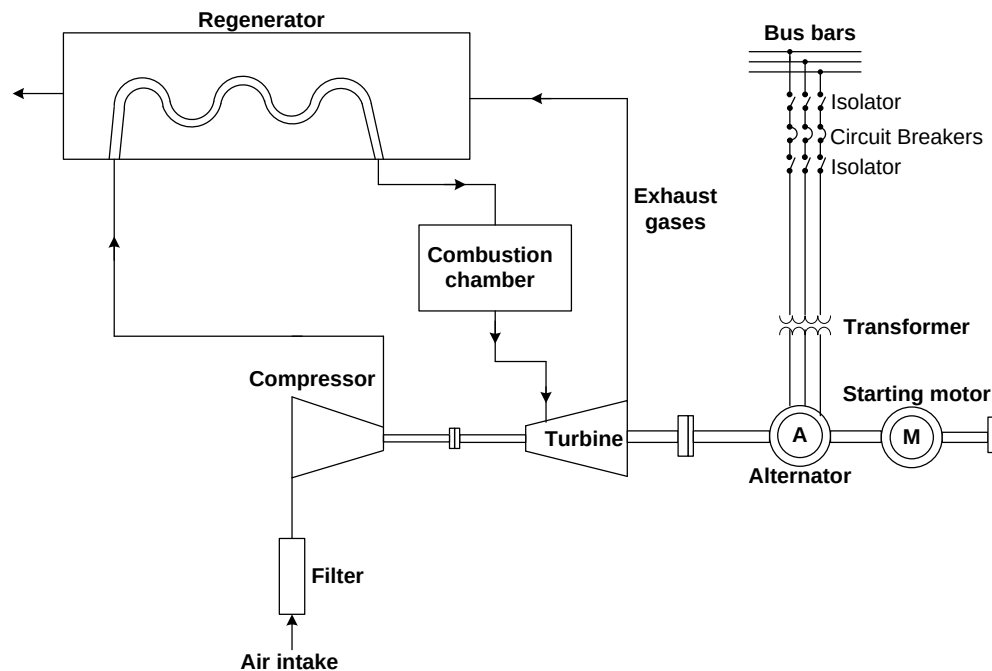


Fig. 8.07. Schematic arrangement of gas turbine power plant

Combustion chamber. The compressed air which is now at high enters the combustion chamber through the regenerator. In the combustion chamber, heat is added to the air by burning oil. The chamber attains very high temperature. The combustion gases are cooled and delivered to the gas turbine.

Gas turbine. The product of combustion comprising of a mixture of gases at high temperature and pressure are passed to the gas turbine making it to turn.

Electrical equipment. The gas turbine is coupled to the alternator. The alternator converts mechanical energy of the turbine into electrical energy. The output from the alternator is fed to the bus-bars through transformer, circuit breakers and isolators.

Starting motor. Before starting the turbine, compressor has to be started. An electrical motor energized by batteries is mounted on the same shaft with the turbine to drive the compressor. Once

the unit is in motion, part of the mechanical power of the turbine is now used to drive the compressor and the motor is switched off.

Advantages

- (i) It can be started quickly from cold conditions.
- (ii) There are no standby losses.
- (iii) It is simple in design.
- (iv) It requires less water since no condenser is used.
- (v) It is much small in size.
- (vi) The initial and operating costs are low
- (vii) Maintenance cost is low.
- (viii) Gas turbines are much simpler in construction and operation than steam turbines.

Disadvantages.

- (i) Before starting the turbine the compressor has to be run by use of external source.
Lowering the overall output of the plant.
- (ii) Low efficiency exhaust gases from the turbine contain sufficient heat.
- (iii) Short life due to very high temperature of combustion.

Review questions

1. With the aid of a diagram describe (i) construction and (ii) Operation of Hydro-electric power plant.
2. Outline the factors to consider when selecting the site for thermal station.
3. With the aid of a single line diagram describe the voltage levels in power supply system.